

Integration Specifications

This document outlines the requirements and standards for integrating external systems, tools, or services with the platform. The goal is to ensure reliable data exchange, consistent performance, and predictable behaviour across all connected components.

1. Integration Scope

- Defines how third-party systems interact with core platform modules.
 - Covers data input, output, authentication, and communication expectations.
 - Ensures that connected tools follow approved formats and processes.
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2. Data Exchange Requirements

- All integrated systems must use approved data structures and naming conventions.
 - Inputs must follow the platform's validation rules to avoid processing errors.
 - Output formats should be predictable and consistent for downstream services.
 - Time stamps, identifiers, and status fields must match the platform's standard formats.
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3. Communication Protocols

- Integrations must use approved communication methods such as REST, Webhooks, or secure messaging channels.
 - Requests must include proper headers, version identifiers, and authentication tokens.
 - Response handling must account for retries, timeouts, and error messaging.
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4. Authentication and Security

- Access requires valid credentials issued by the platform.
- All data must be transmitted through encrypted channels.

- Token management, session limits, and permission scopes must follow defined rules.
 - Integrated systems should implement logging to track access attempts and failed requests.
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5. Performance Expectations

- Requests should meet defined response time targets to prevent bottlenecks.
 - Batch operations or high-volume transactions must follow rate limit guidelines.
 - Systems must handle peak load periods without degrading core platform functions.
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6. Error Handling

- Integrated services should return clear status messages for success, failures, and warnings.
 - Error codes should map to documented categories for easy diagnosis.
 - Recovery steps must be implemented for transient issues like network delays or partial data loss.
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7. Testing and Validation

- Integrations must be tested in a controlled environment before deployment.
 - Validation includes functional checks, performance testing, and security review.
 - Any change to external systems must be retested to ensure compatibility.
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8. Maintenance and Updates

- Integrated systems must remain aligned with platform versioning.
- When API endpoints or data formats change, updates must be applied before the new version goes live.
- Regular audits help ensure the integration remains stable and compliant.